



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI CAUVERY AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

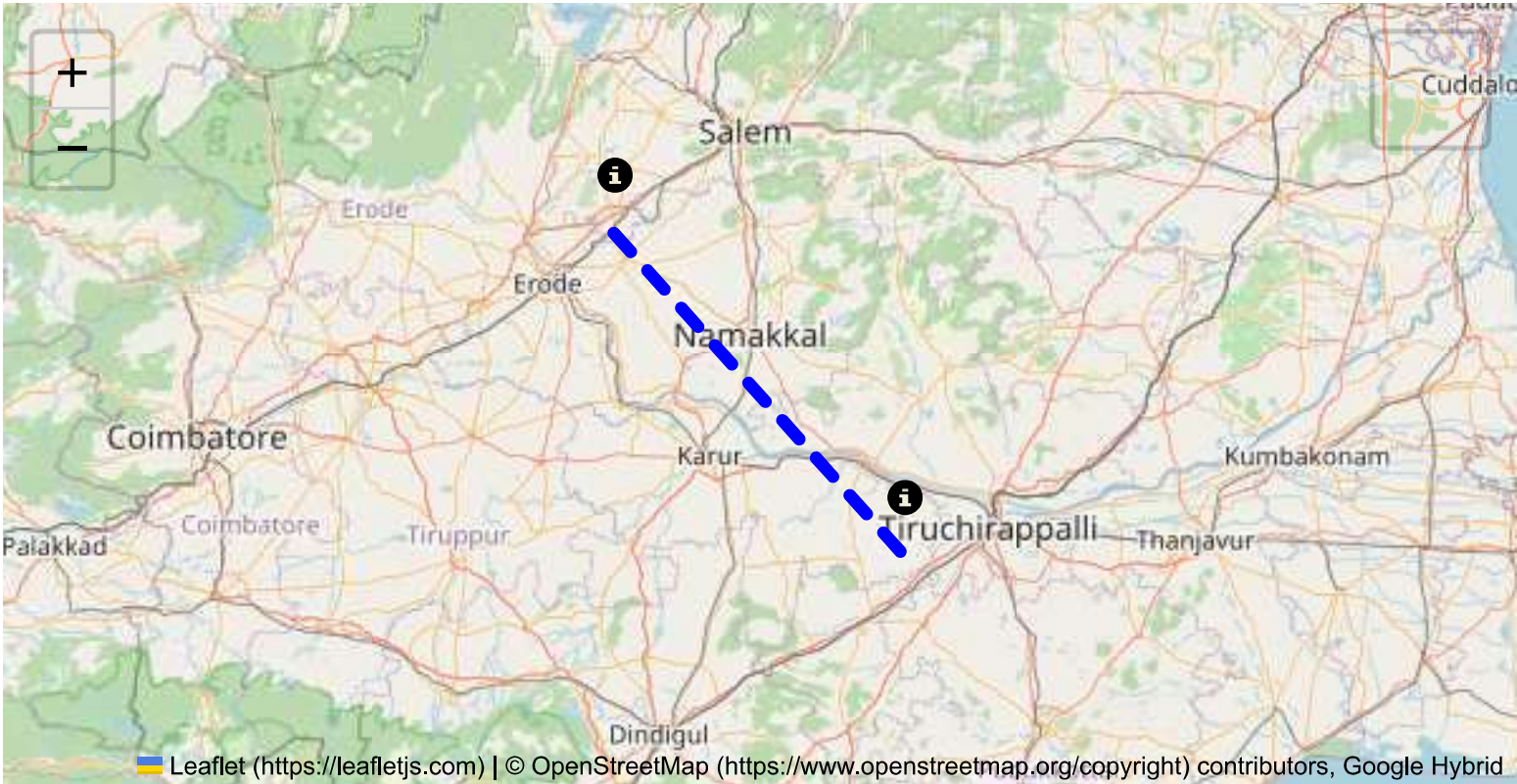
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

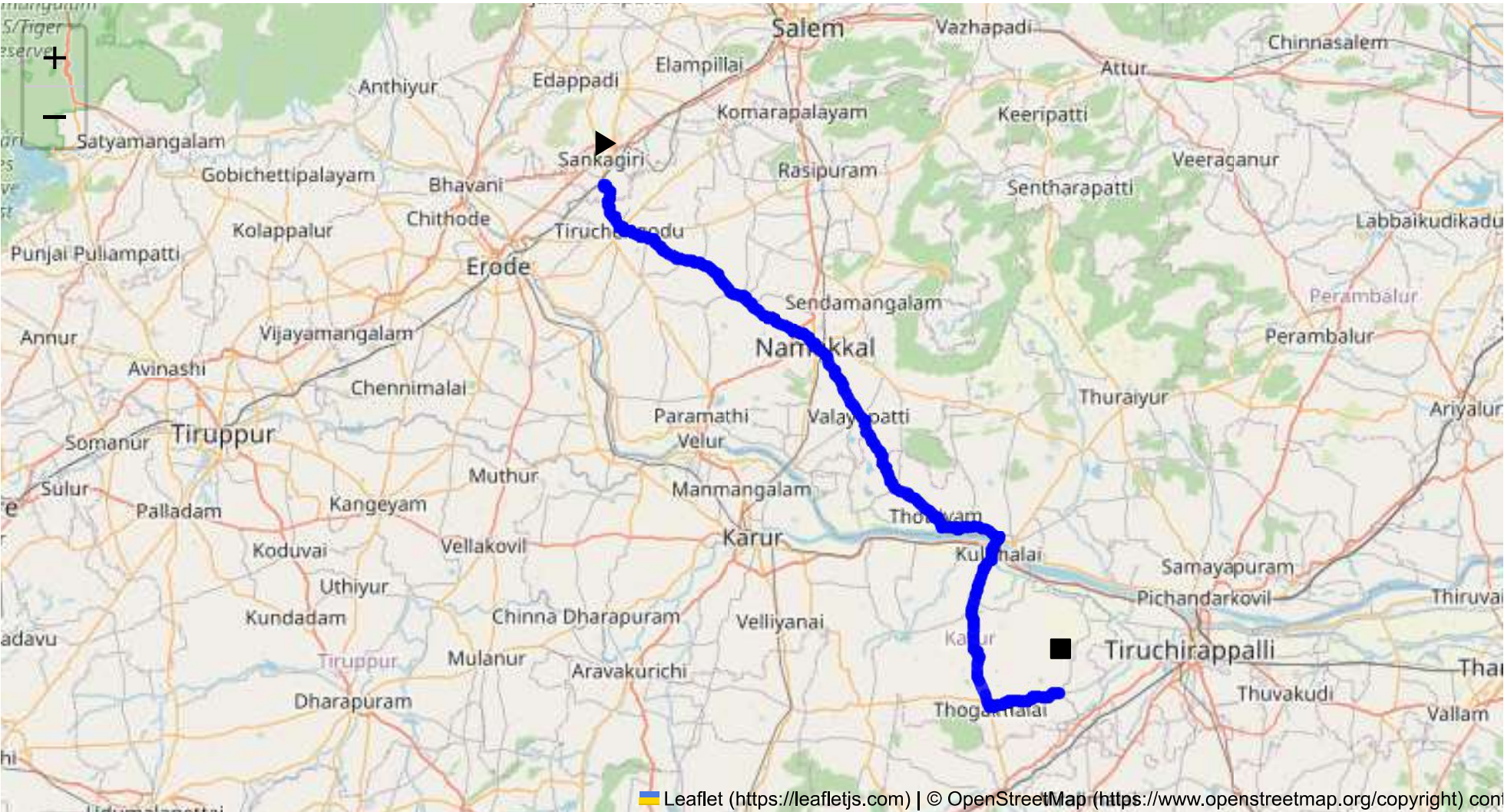
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 127.37 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (10.746645, 78.504433)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Kavalkaranpatti

1. Overview of the Route Map

The route from Sangagiri to Kavalkaranpatti is approximately 127.37 kilometers long. It predominantly follows the NH 44 south to Namakkal, then transfers onto NH 79 before proceeding along several state highways to reach Kavalkaranpatti. The journey typically spans around 2.69 hours for trucks, providing varied road conditions from highways to state and local roads. The route passes through semi-urban and rural areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate, with hot summers, a monsoon season from October to December, and mild winters. The route may encounter:

- Heavy rain during the monsoon, potentially causing waterlogging and reduced visibility.
- High temperatures in summer could affect vehicle performance, especially heavy vehicles.
- Occasional thunderstorms could disrupt travel.

3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Typically, morning (8-10 AM) and evening (5-7 PM)
- **Congestion Areas:** Expect congestion near major town intersections such as Namakkal and Salem due to local and through traffic.
- Road work or local events might cause temporary traffic delays.

4. Assessment of Road Quality and Infrastructure

- **Highways (NH 44 & NH 79):** Generally well-maintained with good surface quality but with occasional potholes that can be hazardous, especially post-monsoon.
- **State Highways:** Varying conditions, some sections might be narrow or poorly lit at night.
- **Bridges:** Essential to verify load limits for trucks, some older bridges might require caution.

5. Suggestions for Alternative Routes for Emergencies

- **Via Namakkal:** Alternate diversion paths through multiple connecting state highways if NH routes are obstructed.
- For traffic diversions, utilizing NH 544 when connecting back towards smaller linking roads could provide a bypass.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Permits required for movement of hazardous materials.
- Restricted movement during peak city hours in urban spots like Salem.
- Adherence to safety and signaling protocols is strictly enforced.

7. Overview of Historical Incidents

- No major recurring incidents, but accidents involving heavy vehicles are generally reported due to speeding or poor conditions in wet weather.
- Check local transport department records for any hazardous material-specific incidents.

8. Environmental Considerations and Sensitive Areas

- Several agricultural zones: Avoid contamination by adhering to spillage and leakage protocols.
- Water bodies near highways; ensure no dumping or leaking chemicals.

9. Communication Coverage

- Generally good on major highways, but be cautious of potential dead zones in rural stretches.
- Equip vehicles with satellite phones or emergency communication devices.

10. Estimated Emergency Response Times

- Highways:** 30-45 minutes average due to good connectivity.
- Rural Areas:** Could extend to 1-2 hours due to reduced accessibility and traffic volume.

11. Overall Summary of Risk Assessment

The route is feasible for heavy vehicles with moderate risk components:

- Weather and Physical Hazards:** Rain and heat pose potential risks, affecting vehicle performance and road conditions.
- Traffic and Infrastructure:** Major towns and poorly maintained state roads can delay travel.
- Ensure compliance with local regulations regarding hazardous materials.
- Communications and emergency preparedness should be optimized, factoring dead zones and potential delays in response times.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.94342, 78.41370	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
5	Turn	Medium	11.38376, 77.89914	30 KM/Hr
6	Turn	High	11.38186, 77.90258	15 KM/Hr
7	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
8	Turn	Medium	11.33419, 78.00855	30 KM/Hr
9	Turn	Medium	11.28043, 78.07413	30 KM/Hr
10	Turn	Medium	11.25724, 78.11087	30 KM/Hr
11	Turn	Medium	11.22407, 78.16640	30 KM/Hr
12	Turn	Medium	11.21962, 78.16777	30 KM/Hr
13	Turn	Medium	11.12309, 78.23255	30 KM/Hr
14	Turn	Medium	10.98680, 78.33772	30 KM/Hr
15	Turn	Medium	10.97288, 78.34365	30 KM/Hr
16	Turn	High	10.96016, 78.42212	15 KM/Hr
17	Turn	Medium	10.96069, 78.42248	30 KM/Hr
18	Turn	Medium	10.96083, 78.42284	30 KM/Hr
19	Blind Spot	Blind Spot	10.96050, 78.42371	10 KM/Hr
20	Turn	Medium	10.94178, 78.41302	30 KM/Hr
21	Turn	Medium	10.80359, 78.39060	30 KM/Hr
22	Turn	High	10.80296, 78.39206	15 KM/Hr
23	Turn	High	10.72750, 78.41006	15 KM/Hr
24	Blind Spot	Blind Spot	10.72727, 78.41137	10 KM/Hr
25	Turn	Medium	10.73148, 78.41289	30 KM/Hr
26	Turn	High	10.73148, 78.41557	15 KM/Hr
27	Turn	Medium	10.73039, 78.41614	30 KM/Hr
28	Turn	High	10.72932, 78.41766	15 KM/Hr
29	Turn	High	10.73304, 78.43557	15 KM/Hr
30	Turn	High	10.73523, 78.43628	15 KM/Hr
31	Turn	Medium	10.73563, 78.44074	30 KM/Hr
32	Turn	Medium	10.73678, 78.44150	30 KM/Hr
33	Turn	Medium	10.73742, 78.45323	30 KM/Hr
34	Turn	Medium	10.73649, 78.45526	30 KM/Hr
35	Turn	Medium	10.73677, 78.46499	30 KM/Hr
36	Turn	Medium	10.73780, 78.46671	30 KM/Hr
37	Turn	High	10.73755, 78.46873	15 KM/Hr
38	Turn	High	10.74164, 78.47191	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
39	Turn	Medium	10.74026, 78.47590	30 KM/Hr
40	Turn	Medium	10.74169, 78.47855	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium
15	hospital	Aruthra Hospital	11.2264666, 78.1631524	30 km/h	Medium
18	clinic	Sri Ram Nursing Home, Namakkal	11.2259986, 78.1631944	30 km/h	Medium
19	clinic	Thangavel Skin Care	11.2253765, 78.1644957	30 km/h	Medium
20	hospital	Rudhra Hospital	11.226569, 78.163216	30 km/h	Medium
21	hospital	Rajendran Children's Hospital	11.2251887, 78.1649142	30 km/h	Medium
22	clinic	Lung Care Centre	11.2276789, 78.1638586	30 km/h	Medium
23	hospital	Vignesh Hospital	11.224818, 78.1621315	30 km/h	Medium
24	clinic	Vishnu Ayurveda Clinic	11.224262, 78.164173	30 km/h	Medium
25	clinic	Dhanvanthiri Physiotheraphy	11.2245256, 78.1638528	30 km/h	Medium
26	hospital	Venkat Hospital	11.2253645, 78.1646159	30 km/h	Medium
27	hospital	Shree Meenakshi Dental Hospital	11.225269, 78.164275	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
28	hospital	Aravinth Hospital	11.2233336, 78.1672805	30 km/h	Medium
29	hospital	Ilamurugan Hospital	11.2248399, 78.166586	30 km/h	Medium
30	hospital	Annai Eye Hospital	11.2241645, 78.1641366	30 km/h	Medium
32	clinic	Kumaran Poly Clinic	11.2233138, 78.1639212	30 km/h	Medium
33	clinic	Chandra Test Tube Baby Centre	11.2241806, 78.1645931	30 km/h	Medium
34	hospital	Aburvaa Ayurveda Hospital	11.2228807, 78.1640453	30 km/h	Medium
35	hospital	Sri Nidhi Hospital, Namakkal	11.2255711, 78.1687291	30 km/h	Medium
36	clinic	Athithya Nursing Home	11.2237726, 78.1686447	30 km/h	Medium
37	hospital	Kathiravan Hospital	11.226651, 78.167351	30 km/h	Medium
38	hospital	Sri Ramana Hospital	11.223503, 78.1685433	30 km/h	Medium
39	hospital	Vanitha Child Care & Skin Hospital	11.2243378, 78.16811	30 km/h	Medium
40	hospital	Sri Dhanvanthries Hospital	11.2241181, 78.1678614	30 km/h	Medium
41	hospital	Nirmala Hospital	11.2179361, 78.1665001	30 km/h	Medium
42	hospital	Surya Hospital, Namakkal	11.2178506, 78.1660375	30 km/h	Medium
43	hospital	Shanthi Hospitals, Namakkal	11.2190661, 78.1685735	30 km/h	Medium
44	hospital	Ganga Hospital	11.21742, 78.170272	30 km/h	Medium
45	hospital	Government Hospital	11.2168924, 78.169068	30 km/h	Medium
47	police	Police Station	11.2185353, 78.1695347	30 km/h	Medium
48	police	Women Police Station	11.2183133, 78.1692677	30 km/h	Medium
50	clinic	Primecare Diabetic Clinic	11.213992, 78.1737091	30 km/h	Medium
51	hospital	Dr. A. Nagarajan Children's Hospital	11.212765, 78.1747068	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
53	hospital	M.M. Hospital	11.2103089, 78.1776545	30 km/h	Medium
54	clinic	Ram Clinic	11.2090922, 78.1790782	30 km/h	Medium
55	clinic	Prasith Clinic	11.208902, 78.180957	30 km/h	Medium
56	hospital	Imayam Speciality Hospital	11.2078899, 78.1797861	30 km/h	Medium
57	clinic	Aburvaa Multi Special Herbal Clinic	11.207568, 78.183483	30 km/h	Medium
58	hospital	Sugam Children's Hospital	11.2060083, 78.1854033	30 km/h	Medium
60	hospital	Government Hospital, Valayapatty	11.1212591, 78.2315962	30 km/h	Medium
63	hospital	Saravana Hospitals	10.9944313, 78.3273838	30 km/h	Medium
64	hospital	Dr. Siva Hospital	10.9877652, 78.3359228	30 km/h	Medium
65	hospital	Dr. Alva Hospital	10.987225, 78.336154	30 km/h	Medium
66	hospital	Ravikumar Hospital	10.988317, 78.336248	30 km/h	Medium
67	police	Thottiyam Police station	10.9874415, 78.3363001	30 km/h	Medium
68	hospital	Government Hospital, Thottiyam	10.98388, 78.338648	30 km/h	Medium
69	hospital	Government Hospital	10.9428055, 78.4146362	30 km/h	Medium
70	hospital	T M R Hospital	10.9429825, 78.4158242	30 km/h	Medium
71	hospital	Thogamalai, Government Hospital	10.7268532, 78.4097304	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
12	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
16	school	Government Elementary School	11.224555, 78.163967	30 km/h	Medium
17	school	Government Secondary School	11.2244426, 78.1647533	30 km/h	Medium
31	school	Government Girls School	11.2238954, 78.1662583	30 km/h	Medium
46	school	Government Girls Higher Secondary School	11.2179186, 78.1672383	30 km/h	Medium
49	school	Selvam Matriculation Higher Secondary School	11.2124286, 78.1770824	30 km/h	Medium
52	college	Namakkal Kavignar Ramalingam Government Arts College for Women	11.2125137, 78.1724654	30 km/h	Medium
59	college	King college	11.1692622, 78.2037212	30 km/h	Medium
61	school	Government Higher Secondary School	11.0234395, 78.2859835	30 km/h	Medium
62	college	Vetri vinayaga institution	11.013503, 78.3072053	30 km/h	Medium

Route Photos of Risky Spots